



GITHUB

DATA-DRIVEN GRAPH INFERENCE FOR INFRASTRUCTURE NETWORKS

Estéban Nocet-Binois, supervised by Jürgen Hackl

Department of Civil & Environmental Engineering | Princeton University, NJ, USA



MOTIVATION

The objective is to learn a basis that captures the dependencies and global modes of variation (called graph harmonics, or *frequencies*) present in the data. Analyzing how system states (like frequency deviations) decompose into these modes reveals which patterns are most likely to persist or amplify after a disturbance. Thus, given a *meaningful* basis, we can analyze how signals propagate, diffuse, or concentrate, and how oscillations or disturbances (such as power outages) manifest across the entire grid.



NOTATIONS

Graph Laplacian

- The Laplacian $L \in \mathbb{R}^{n \times n}$ encodes how values at each node deviate from its neighborhood. We denote the eigendecomposition:

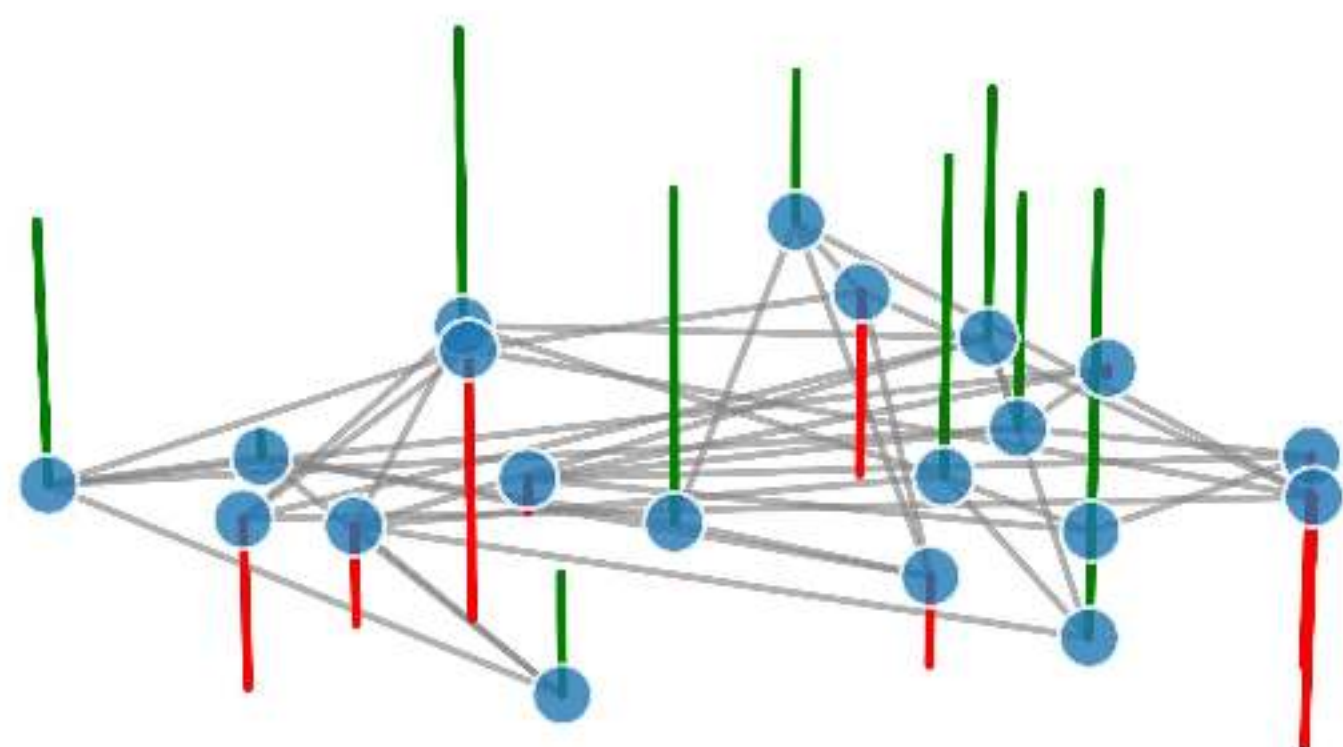
$$L = U\Lambda U^\top, \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$$

- We note the pseudoinverse:

$$L^\dagger = \sum_{k=1}^{n-1} \frac{1}{\lambda_k} u_k u_k^\top$$

Signal Matrix

- A graph signal is a n -valued tensor, collecting states across all nodes.



- Collection $X \in \mathbb{R}^{m \times n}$ of m signals (rows) over n nodes (columns)
- Sample covariance:

$$\Sigma = \frac{1}{m} X^\top X$$

The objective is, given a series of graphs signals, and optional adjacencies, to infer the dependency structure as edge weights.

LAPLACIAN LEARNING

$L^\dagger$ encodes graph topology via $\mathbb{E}[\Sigma]$.

Edge Interpretation

- $\Omega = L$: Precision matrix
- $\Omega_{i,j} = -W_{i,j}$ (partial correlations)
- $\Omega_{i,i} = D_i$ (degree)

Conditional Independence

$$x_i \perp x_j \mid X_{\setminus\{i,j\}} \iff L_{i,j} = 0$$

Maximum Likelihood

$$\max_{L \in \mathcal{L}} \log \det(L) - \text{tr}(L\Sigma)$$

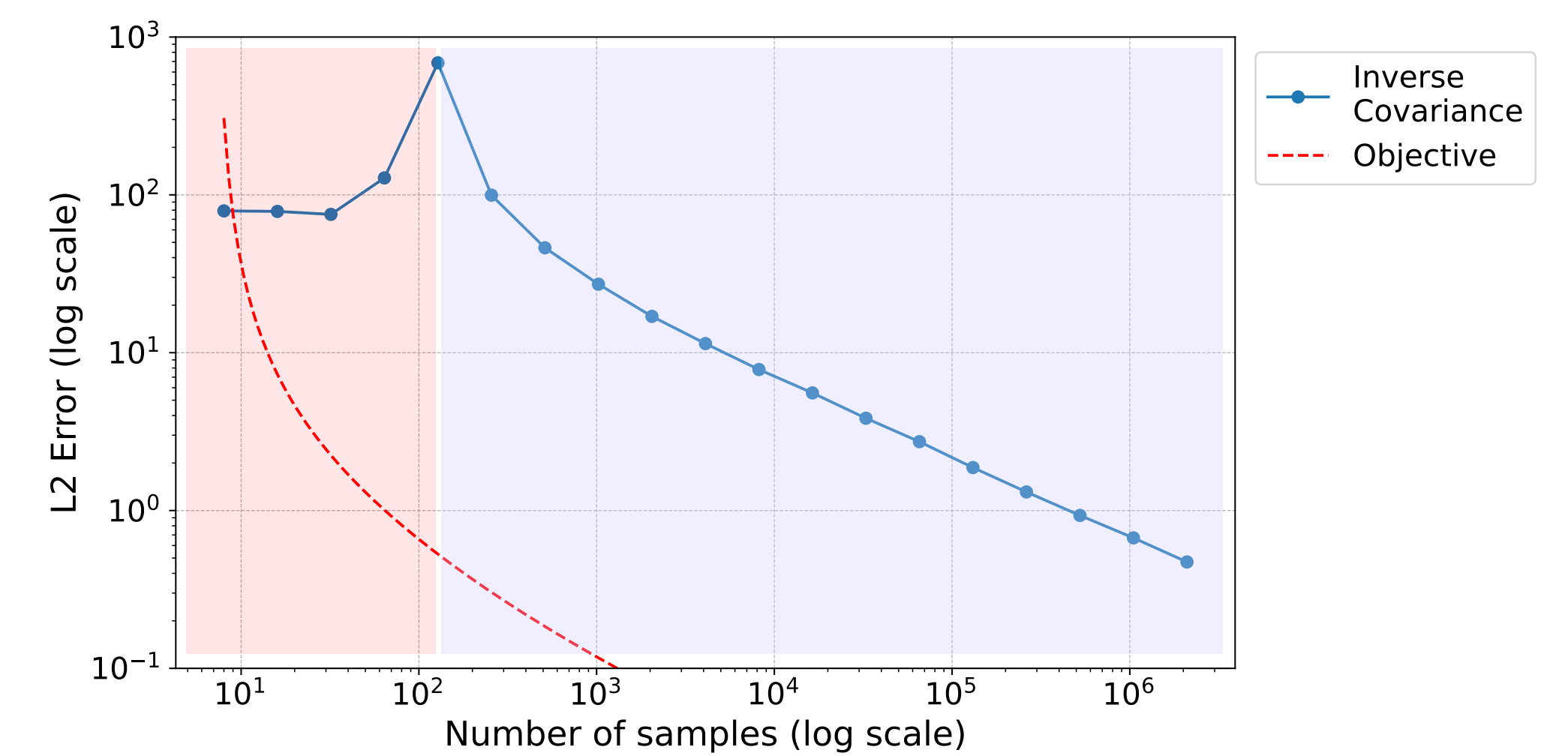
Promotes large eigenvalues (smooth surrogate for rank)
Penalizes deviation from covariance

$$\mathcal{L} = \{L : L \succeq 0, L1 = 0, L_{i \neq j} \leq 0\}$$

Off-diagonal entries are non-positive
Rows sum to zero (centering)
 L is positive semidefinite

The determinant of a rank deficient (square) matrix is always zero, which happens:

- When Σ is rank-deficient ($k < n$), as we minimize $\text{tr}(L\Sigma)$.
- When enforcing Laplacian constraints (hence singularity by design).



Sparsity Penalty

$$\min_{\Theta \succ 0} \left\{ \text{tr}(\Theta\Sigma) - \log |\Theta| + \beta \sum_{i \neq j} \rho(\Theta_{ij}) \right\}$$

- The sparse penalty function $\rho(\cdot)$ (e.g. ℓ_1 , SCAD, MCP or LCP) prevents edge weight shrinkage and overfitting.
- It also helps with stability by preventing Θ from becoming nearly singular.

Ridge Regularization

- We learn the shifted precision $\Theta + \delta I \succ 0$ under the constraint $\Theta \in \mathcal{L}$.

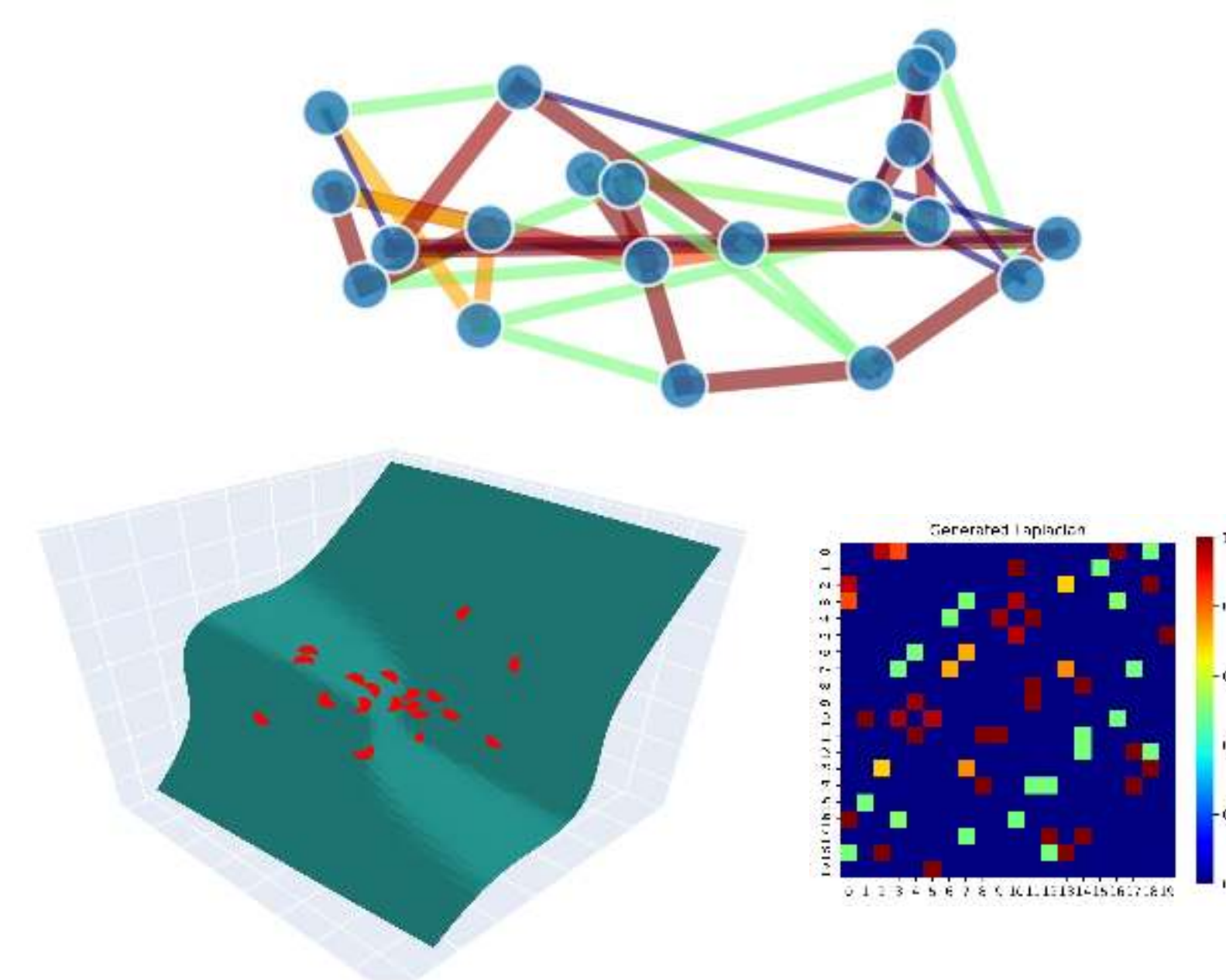
Spectral Relaxation

$$\min_{L \in \mathcal{L}} \left\{ \text{tr}(L\Sigma) + \alpha_1 \|L\|_{F,\text{off}}^2 - \alpha_2 \sum_i \log(L_{ii}) \right\}$$

- Strongly convex due to Frobenius norm, while implicitly enforcing spectral properties.
- Stabilizes estimation when Σ is rank-deficient.

MANIFOLD STRUCTURE

- As $n \rightarrow \infty$ and bandwidth $\epsilon \rightarrow 0$, $L \rightarrow \Delta$ (Laplace-Beltrami) under uniform sampling.



- Dirichlet energy, whose minimization solves $\Delta f = 0$:

$$\mathcal{E}(f) = \int_{\mathcal{M}} \|\nabla f\|^2 \leftrightarrow x^\top Lx$$

- The Laplacian eigenvalues and eigenvectors provide a Fourier-like basis for graph signals, enabling frequency analysis.

The task shifts from learning a graph Laplacian that reflects data covariance, to identifying low-dimensional manifolds that preserve local neighborhoods.

- Diffusion maps learn a set of orthonormal basis functions—the eigenvectors of the diffusion operator H_t :

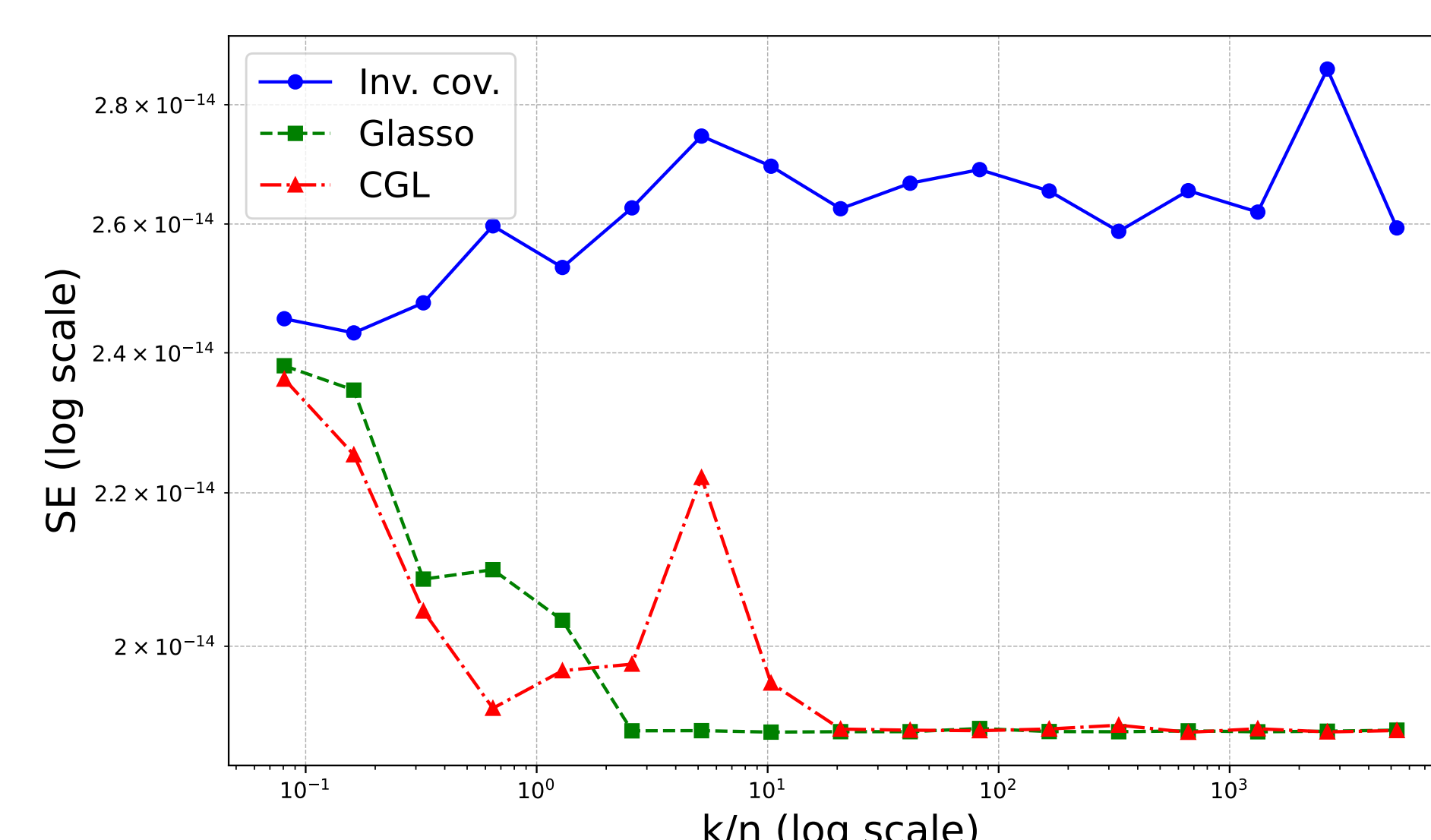
$$\frac{\partial f}{\partial t} = -L f \iff f(t) = \underbrace{e^{-tL}}_{H_t} f_0$$

ALIGNMENT RESULTS

Given two symmetric matrices $L, \hat{L}$ and their projection matrices $P, \hat{P}$, the subspace error is:

$$SE = \|P - \hat{P}\|_F, \quad P = UU^\top$$

This measures how different the two subspaces (spanned by the bases $U, \hat{U}$ in which the Laplacians are diagonal) are: it is zero if they span the same subspace, and larger if their bases are more misaligned.



FUTURE WORK

Interference and Non-Linearity

- Can we generalize diffusion-based approaches to distortion-based ones?
- Can we go beyond i.i.d. time series?
- Can we have theoretical guarantees for nonlinear dynamics?

Application to Infrastructures

- Water flow in water distribution systems (WDSs): Can a data-driven Laplacian help producing signals that are compressible or smooth?
- Water quality and propagation in sensor networks (over WDSs): How can we infer contaminant transport from partially observed signals?
- Power grid stability: How can we capture ways in which disturbances propagate across a grid?

- [1] M. Belkin and P. Niyogi. Towards a theoretical foundation for Laplacian-based manifold methods. *J. Comput. Syst. Sci.*, 74(8):1289–1308.
- [2] R. R. Coifman and S. Lafon. Diffusion maps. *Appl. Comput. Harmon. Anal.*, 21(1):5–30.
- [3] A. P. Dempster. Covariance selection. *Biometrics*, 28(1):157–175.
- [4] X. Dong, D. Thanou, P. Frossard, and P. Vandergheynst. Learning Laplacian matrix in smooth graph signal representations. *IEEE Trans. Signal Process.*, 64(23):6160–6173.
- [5] X. Dong, D. Thanou, M. Rabbat, and P. Frossard. Learning graphs from data: A signal representation perspective. *IEEE Signal Process. Mag.*, 36(3):44–63.
- [6] H. E. Egilemez, E. Pavez, and A. Ortega. Graph learning from data under Laplacian and structural constraints. *IEEE J. Sel. Top. Signal Process.*, 11(6):825–841.
- [7] M. Erbar. The heat equation on manifolds as a gradient flow in the Wasserstein space. *Ann. Inst. H. Poincaré Probab. Statist.*, 46(1).
- [8] J. Friedman, T. Hastie, and R. Tibshirani. Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9(3):432–441.
- [9] A. Heimowitz and Y. C. Eldar. A unified view of diffusion maps and signal processing on graphs. In *SampTA 2017*, pages 308–312. IEEE.
- [10] K. Schultz, M. Villafañe-Delgado, E. P. Reilly, G. M. Hwang, and A. Saksena. Graph signal processing for infrastructure resilience: Suitability and future directions. In *2020 Resilience Week (RWS)*, pages 64–70.
- [11] A. Singer. From graph to manifold Laplacian: The convergence rate. *Appl. Comput. Harmon. Anal.*, 21(1):128–134.
- [12] R. Talmon, I. Cohen, S. Gannot, and R. R. Coifman. Diffusion maps for signal processing. *IEEE Signal Process. Mag.*, 30(4):75–86.
- [13] L. Xia, X. Huang, G. Wang, and T. Wu. Positive-definite sparse precision matrix estimation. *APM*, 07(01):21–30.